

EXECUTIVE SUMMARY

FERRY CAPACITY UTILIZATION & OPERATIONAL EFFICIENCY ANALYTICS SYSTEM

This project analyzes the Toronto Island Ferry Ticket Dataset (2015–2025) to evaluate ferry capacity utilization and operational efficiency. The analysis is based on ticket sales and redemption records collected at 15-minute intervals. The primary objective is to identify peak demand periods, idle operating windows, and utilization patterns that can support better operational planning.

The dataset was preprocessed by converting timestamps into datetime format, checking for missing values and duplicates, and creating additional analytical features such as Total Activity Load, Redemption Pressure Ratio, Operational Load Index, and Idle Capacity Indicator.

Exploratory Data Analysis (EDA) was performed to understand passenger activity across different time intervals, weekdays, weekends, and seasons. Time-series analysis was conducted using daily, hourly, and monthly aggregations to identify trends in ferry operations.

Several Key Performance Indicators (KPIs), including Capacity Utilization Ratio, Congestion Pressure Index, Idle Capacity Percentage, Peak Strain Duration, and Operational Variability Score, were calculated to measure operational efficiency. These KPIs provide valuable insights into ferry performance and help identify opportunities for improving scheduling and resource allocation.

An interactive Streamlit dashboard was developed to visualize operational metrics, activity trends, and KPI summaries. The dashboard enables users to explore ferry performance through charts and analytics, supporting data-driven decision-making.

The project concludes that historical ticket data can be effectively used to improve ferry scheduling, optimize resource utilization, reduce operational inefficiencies, and enhance passenger service. The proposed analytics framework provides practical insights that can support long-term planning and more efficient public transportation operations.